

Logistic Regression

- Used in supervised learning where output labels are either 1 or 0, for binary classification.

Given a feature vector x , we want:

$$\hat{y} = P(y=1|x)$$

↳ probability of y being 1 given x

Parameters of logistic regression:

$$w \in \mathbb{R}^n \quad b \in \mathbb{R}$$

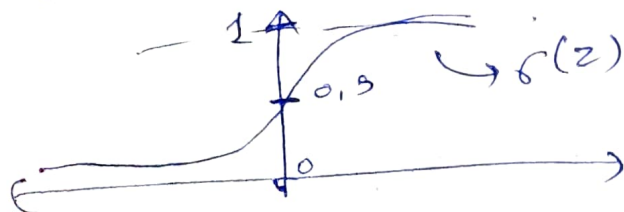
Output: $\hat{y}$ = Some function which gives a value b/w 0 and 1

General form

~~Let's say~~ $\hat{y} = w^T x + b$ → This doesn't work for probability, so we use sigmoid.
input feature vector

• Sigmoid:

- $\sigma(z)$ sigmoid of a function z is plotted as:



It keeps the value b/w 0 & 1. just what we want

Formula for σ is:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

In logistic regression, the job is to learn w & b such that $\hat{y}$ is as close to 1 as possible

Note: It is easier in practice to keep b & w separate.

In some contents or textbooks and stuff, they will represent

$\hat{y}$ as:

$$\hat{y} = \sigma(\phi^T x) \quad \text{take } x_0 = 1 \quad x \in \mathbb{R}^{n_x+1} \quad \rightarrow \text{extra feature}$$

where

$$\phi = \begin{bmatrix} \phi_0 \\ \phi_1 \\ \vdots \\ \phi_n \end{bmatrix} \quad \left. \begin{array}{l} \rightarrow b \\ \rightarrow w \end{array} \right\} \text{ here } \phi_0 \text{ acts as } b \text{ and rest of } \phi \text{ is } w$$